

6. ULTREASONIC

```
const int trigPin = 12;

const int echoPin = 11;

const int LED_PIN = 10;

const int distance_threshold = 20;

float duration_us, distance_cm;

void setup() {

    Serial.begin(9600);

    pinMode(trigPin, OUTPUT);

    pinMode(echoPin, INPUT);

    pinMode(LED_PIN, OUTPUT);

}

void loop() {

    digitalWrite(trigPin, HIGH);

    delayMicroseconds(10);

    digitalWrite(trigPin, LOW);

    duration_us = pulseIn(echoPin, HIGH);

    distance_cm = 0.017 * duration_us;

    if

(distance_cm < distance_threshold)    digitalWrite(LED_PIN, HIGH);

    else

        digitalWrite(LED_PIN, LOW);

    Serial.print("distance:");

    Serial.print(distance_cm);

    Serial.println("cm");

    delay(500);

    Serial.print("duration");

    Serial.print(duration_us);
```

